

Mitigating Time-Shift Errors in CGM-based Glucose Forecasting and Hypoglycemia Event Prediction

Bachelor Thesis

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submitted by

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Abstract

Planned Content:

- **Problem.** Deep learning models trained with mean squared error (MSE) on univariate CGM data produce glucose predictions that are systematically delayed by about 50 to 55 minutes. This time-shift artefact causes hypoglycemia events (glucose below 70 mg/dL) to be detected too late or not at all.
- **Methods.** Three objectives are evaluated in a controlled ablation study. Only one factor changes per objective; all other settings are identical to the baseline.
 - Objective 1: offset-aware loss functions, specifically Bounded-Lag ($D=3$ steps, ± 15 min) and Soft-DTW ($\gamma=1.0$)
 - Objective 2: multi-step forecasting with simultaneous prediction of all $H=12$ steps (DIRMO strategy)
 - Objective 3: event-centric evaluation with a τ -sweep ($\tau \in \{1, \dots, 6\}$ steps) and a direct binary event classifier (BCE loss)
- **Forecasting results.** Multi-step MSE achieves the lowest lag_rmse of all tested methods (LSTM: 20.74 mg/dL, PatchTST: 14.53 mg/dL). Loss function changes alone do not reduce the time-shift artefact.
- **Event detection results.** The LSTM binary classifier achieves $F1 = 0.334$, a 62-fold improvement over forecast-derived detection ($F1 = 0.005$). Recall = 0.693 means 69 % of hypoglycemia events are correctly detected. The PatchTST classifier achieves only $F1 = 0.095$ due to very low precision caused by its Reversible Instance Normalization removing absolute glucose level information.
- **Conclusion.** Forecasting and event detection work best as two separate tasks. The recommended setup is a multi-step MSE model for glucose trajectory prediction and an LSTM binary classifier for hypoglycemia early warning.

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Planned Content:

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Chapter 1

Introduction

This chapter motivates the presented research, introduces the central problem of the systematic time-shift anomaly in machine-learned glucose predictions, and defines the three core research objectives. Section 1.1 describes the underlying clinical context of Type 1 diabetes. Section 1.2 explains the time-shift artefact that motivates the ablation study. Section 1.3 defines the scope of the evaluation and states the research questions. Section 1.4 summarizes the primary scientific contributions, and Section 1.5 details the overall structural organization of the thesis.

1.1 Clinical Motivation

Planned Content:

- Characterization of Type 1 Diabetes Mellitus as an autoimmune condition involving the complete destruction of endogenous insulin production which necessitates lifelong external insulin administration.
- Presentation of global epidemiological data using the IDF Diabetes Atlas to highlight the expanding relevance of the condition and the persistent high risk of acute complications.
- Explanation of Continuous Glucose Monitoring technology as a continuous sensor-based measurement tool that provides the foundational input signal for modern automated insulin delivery loops.
- Definition of hypoglycemia thresholds and the critical clinical need for early warning mechanisms due to the physiological pharmacodynamic delays of insulin.

- Description of Figure 1.1: Visual presentation of a real hypoglycemia episode where the interstitial glucose concentration crosses the critical threshold line during a steep downward trajectory.

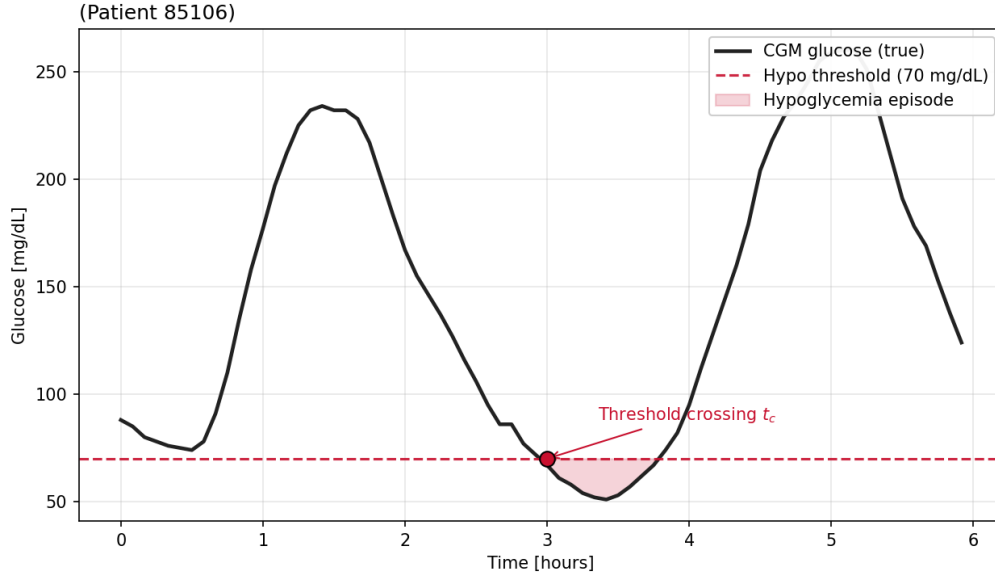


Figure 1.1: Typical hypoglycemia episode in the CGM signal of a Type 1 diabetes patient. The glucose value drops below the clinical threshold of 70 mg/dL. A preventive measure must be initiated already during the downward phase, not only after the threshold has been crossed, which motivates the need for accurate early-warning systems addressed in this thesis.

1.2 The Time-Shift Problem in Glucose Forecasting

Planned Content:

- Definition of the systematic time-shift anomaly where standard regression models optimize point-wise criteria but produce predictions delayed by approximately 50 to 55 minutes.
- Analysis of the mathematical root cause stemming from the high lag-1 autocorrelation of the univariate glucose signal which induces models to mimic historical states rather than anticipating future trends.

- Explanation of the severe clinical consequences of this delay because delayed threshold crossings render forecast-based event detection mechanisms ineffective.
- Introduction of the `lag_rmse` diagnostic metric as a specialized evaluation tool designed to quantify the specific error component caused by temporal misalignment.
- Description of Figure 1.2: Visual comparison between the actual glucose trajectory and a delayed model forecast to demonstrate the shape-preserving but temporally shifted nature of the prediction.

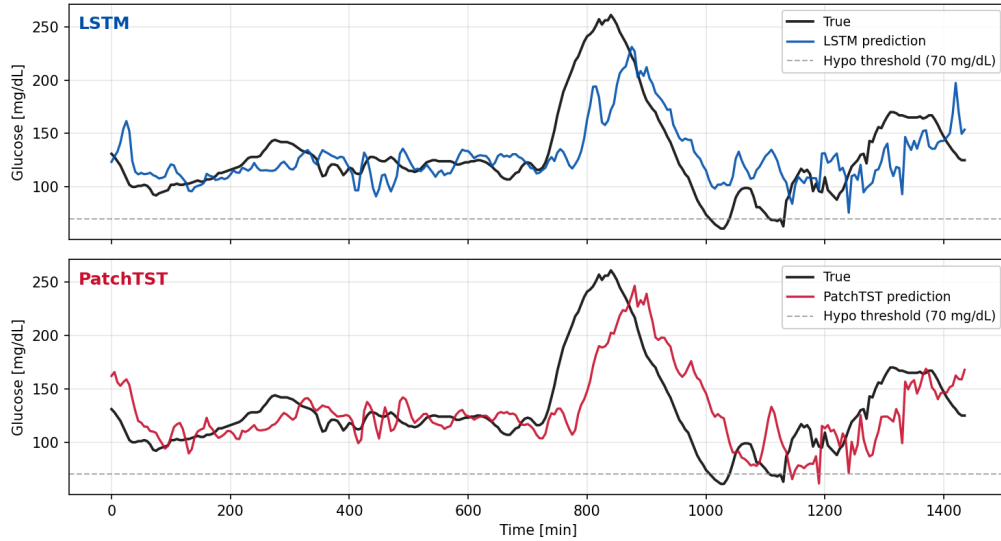


Figure 1.2: Illustration of the time-shift artefact in MSE-trained univariate glucose forecasting models. Both the LSTM (top) and PatchTST (bottom) predictions follow the true glucose trajectory closely in shape but are delayed by approximately 55 minutes. Hypoglycemia threshold crossings are therefore detected too late, or not at all, which motivates the `lag_rmse` diagnostic metric introduced in this thesis.

1.3 Scope and Research Questions

Planned Content:

- Overview of the T1DATA clinical dataset consisting of continuous glucose monitoring records from 36 anonymized patients over multiple weeks.

- Justification for the strictly univariate modeling approach to deliberately isolate the temporal predictive capacity of the glucose signal itself without confounding external inputs.
- Selection rationale for the two evaluated network architectures: the recurrent Long Short-Term Memory network and the patch-based PatchTST Transformer model.
- Description of the controlled ablation design framework where exactly one configuration variable is altered per objective while all underlying hyperparameters remain constant.
- Formal presentation of the three central research questions regarding the impact of offset-aware loss functions, multi-step output strategies, and direct binary event classification.

1.4 Contributions

Planned Content:

- Execution of a highly standardized ablation study comprising multiple training methodologies evaluated uniformly across a substantial cohort of 36 independent patients.
- Methodological incorporation of the lag_rmse metric to separate structural shape errors from pure temporal misalignment components within the total error.
- Formulation of the tolerance-sweep evaluation protocol to systematically map the detection capabilities of different forecasting models across multiple distinct time windows.
- Demonstration of the architectural performance inversion where specific transformer properties excel at trajectory forecasting while recurrent structures show superior classification alignment.

1.5 Thesis Structure

Planned Content:

- Structural outline of Chapter 2 (Background and Related Work) establishing the technical foundations of continuous glucose monitoring, sequential deep learning architectures, offset-aware loss formulations, and evaluation metrics, followed by a critical review of related work.
- Roadmap for Chapter 3 (Experimental Design and Training Objectives) detailing the mathematical problem formulation, data partitioning protocols, the controlled ablation framework, and the specific configuration of each of the three research objectives.
- Overview of Chapter 4 (Empirical Results of the Ablation Study) summarizing the baseline results, the individual outcomes for all three objectives, and the comprehensive comparative analysis across architectures and training strategies.
- Description of Chapter 5 (Conclusions and Future Work) outlining the final analytical summary, identified methodological limitations, and strategic directions for subsequent research.

Chapter 2

Background and Related Work

This chapter provides the technical foundation for the thesis. Section 2.1 describes continuous glucose monitoring technology and the properties of the clinical dataset. Section 2.2 introduces the deep learning architectures utilized for sequential modeling. Section 2.3 details the mathematical formulations of the evaluated loss functions. Section 2.4 defines the point-wise and event-centric evaluation metrics, and Section 2.5 discusses related work in glucose forecasting and time-shift mitigation.

2.1 Continuous Glucose Monitoring

2.1.1 CGM Technology and Sensors

Planned Content:

- Explanation of the electrochemical operational mechanics of continuous glucose monitoring sensors alongside the physiological delay between blood and interstitial glucose compartments.
- Technical specifications of the Dexcom G6 sensor platform highlighting the sensor lifespan and wireless data transmission protocols to external receivers.
- Clinical utility of real-time monitoring streams as critical input features for predictive alerts and automated insulin delivery control algorithms.
- Structural modeling difficulties arising from signal noise and sensor calibration drifts combined with a high lag-1 autocorrelation exceeding 0.98.

2.1.2 The T1DATA Clinical Dataset

Planned Content:

- Description of the internal T1DATA clinical trial dataset comprising continuous tracking records from 36 patients with Type 1 diabetes collected over multiple weeks.
- Description of Table 2.1: high-level cohort statistics summarizing the patient population, sampling rate, total record count, and the prevalence of hypoglycemia events.
- Rationale for training individual specialized models per patient to capture idiosyncratic metabolic dynamics instead of striving for cross patient generalization.
- Characterization of the severe class imbalance inherent to the dataset because true hypoglycemia onset windows constitute a small minority of the total observations.
- Description of Figure 2.1: visual presentation of the class imbalance, motivating both the F2 metric in Section 2.4.3 and the inverse-frequency weighted BCE loss used by the direct classifier in Section 3.5.2.
- Forward reference: the technical train/validation/test splits and per-patient normalization choices are described in Section 3.2.1.

Table 2.1: High-level summary statistics of the T1DATA cohort used throughout this thesis. The table characterizes the dataset along the dimensions most relevant to the modeling decisions made later: the number of independent patients motivates the per-patient training strategy, while the hypoglycemia prevalence motivates the class-imbalanced evaluation framework.

Property	Value
Number of patients (after exclusion)	36
Patients excluded (no hypo events in train)	TODO
Sampling interval	5 min
Mean records per patient	TODO
Total samples (test split, all patients)	TODO
Mean glucose $\pm$ SD (mg/dL)	TODO $\pm$ TODO
Hypoglycemia prevalence (% of samples below 70 mg/dL)	TODO %
Lag-1 autocorrelation of glucose signal	> 0.98

*Placeholder for `img/class_imbalance.png`.
Bar chart of positive vs. negative windows per
patient, plus cohort-level aggregate.*

Figure 2.1: Distribution of positive (hypoglycemia within the forecast horizon) and negative windows across the 36 patients of the T1DATA cohort. The severe minority status of positive windows motivates the use of recall-weighted metrics (F2) and the inverse-frequency weighting in the BCE loss of the direct classifier.

2.2 Deep Learning for Time Series Forecasting

2.2.1 Long Short-Term Memory Networks

Planned Content:

- Theoretical foundation of recurrent neural networks for sequential data and the resolution of the vanishing gradient problem through specialized internal cell states.
- Mathematical mechanics of the input forget and output gating mechanisms that regulate the flow and retention of long term sequence information.
- Specification of the chosen baseline recurrent network topology incorporating two stacked layers with a hidden size of 256 neurons and a linear projection head.
- Description of Figure 2.1: Architectural diagram mapping the processing flow from a 24 step lookback sequence to the respective forecasting or classification output heads.

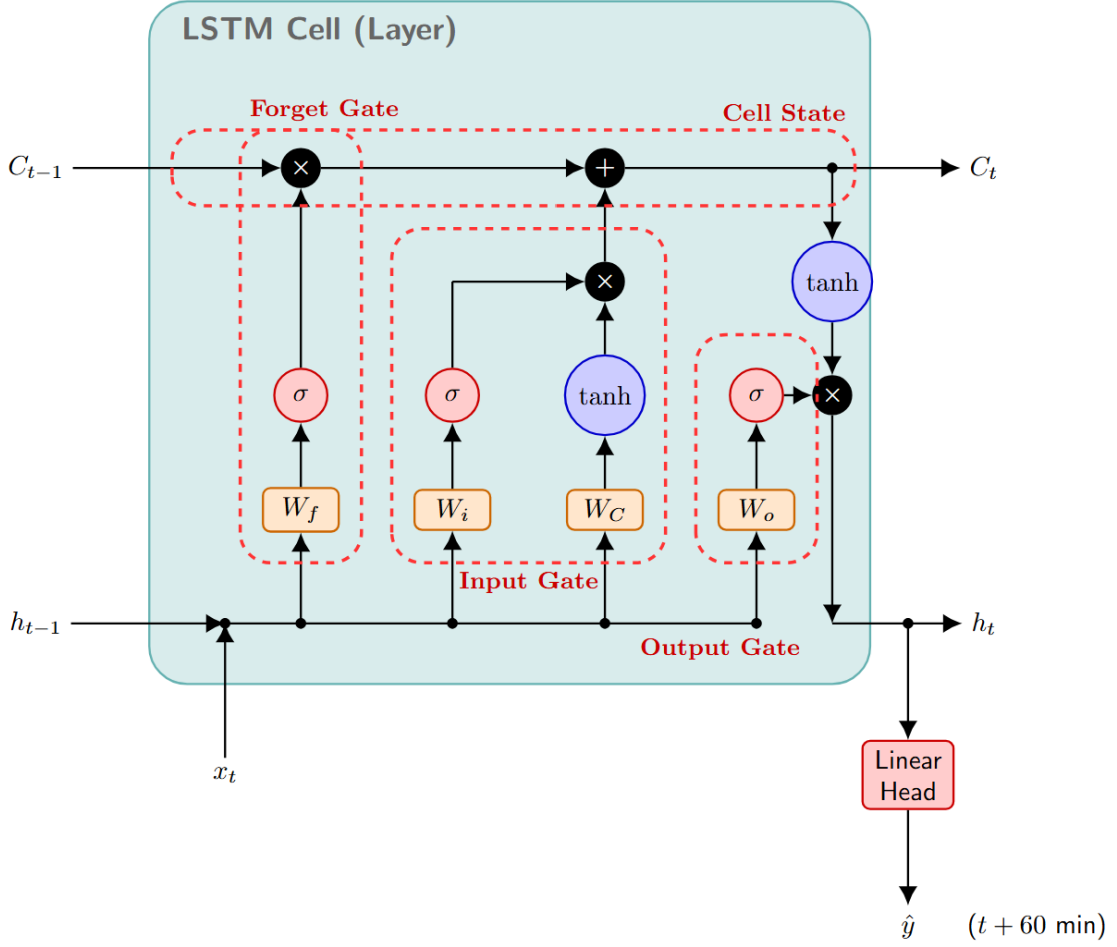


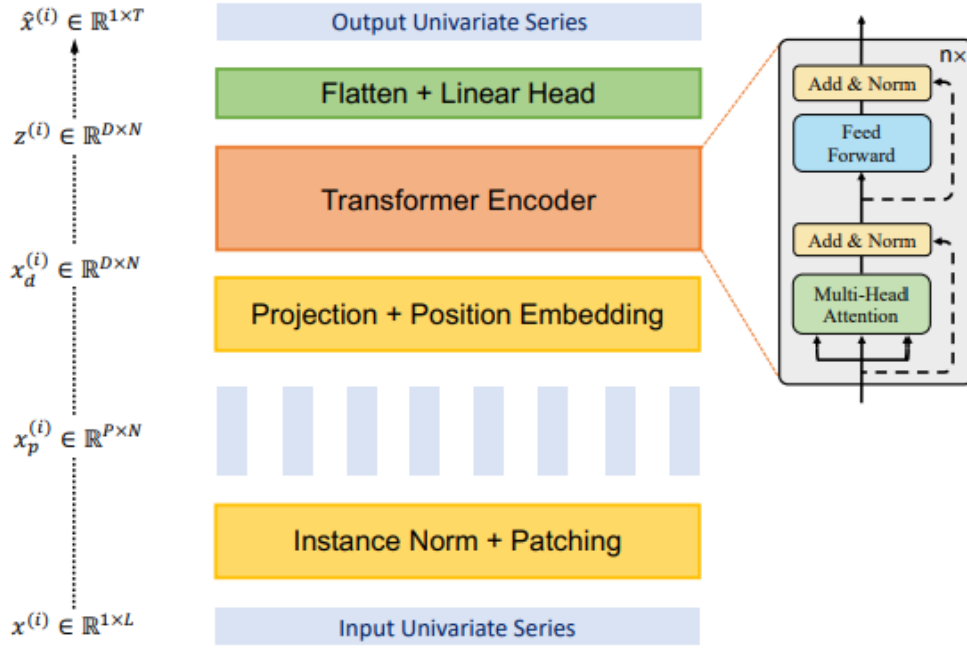
Figure 2.2: Architecture of the LSTM model used as the recurrent baseline in this thesis. The input is a 24-step lookahead window representing 120 minutes of CGM data. A two-layer stack with hidden size 256 feeds into a linear projection head whose output role differs by task: a single scalar for regression (60-minute forecast), a sequence of $H = 12$ scalars for multi-step regression, or a single sigmoid neuron for binary hypoglycemia classification.

2.2.2 PatchTST

Planned Content:

- Review of self-attention mechanisms in time series analysis and the computational efficiency limitations of executing point wise attention on long sequences.
- Core design of the PatchTST architecture where segmentation into overlapping patches reduces input tokens while enhancing local semantic cohesion.

- Mathematical framework of Reversible Instance Normalization utilized to stabilize training by removing instance specific mean and variance transformations.
- Analysis of the structural consequence of instance normalization which strips absolute level info and compromises classification against static boundaries.
- Description of Figure 2.2: Flowchart illustrating the patch extraction process followed by transformer processing and the final inversion of the normalization layer.



(b) Transformer Backbone (Supervised)

Figure 2.3: Architecture of the PatchTST model after Nie et al. [1], adapted to the CGM forecasting setting of this thesis. The 24-step input sequence is segmented into 3 overlapping patches of length 12 with stride 6 before processing. Reversible Instance Normalization (RevIN) [2] removes the instance-specific mean and variance prior to encoding and re-applies the scaling at the output head. The implication of this normalization for the direct binary classifier is discussed in Section 4.4.2.

2.3 Loss Functions for Sequential Regression

2.3.1 Mean Squared Error

Planned Content:

- Definition of the conventional mean squared error loss function which applies a symmetric quadratic penalty to point wise differences.
- Limitation of point wise criteria in time series settings where temporal delays are penalized heavily despite identical shape matching.

2.3.2 Soft-DTW: Differentiable Dynamic Time Warping

Planned Content:

- Conceptual review of traditional dynamic time warping for elastic alignments and the formulation of Soft-DTW as a differentiable soft minimum function.
- Impact of the smoothing regularization parameter on training stability and the optimization of global structural shape patterns.
- Theoretical trade-off where structural shape matching de-prioritizes exact amplitude tracking which potentially increases absolute deviation metrics.

2.3.3 Bounded-Lag Loss

Planned Content:

- Functional introduction of the Bounded-Lag objective designed to eliminate penalties for temporal misalignments falling within a predefined local neighborhood.
- Algorithmic description of the localized minimization process where gradients flow exclusively through the single best matching temporal offset step.

- Constraints of the localized optimization framework when the boundary parameter is narrower than the actual systematic delay observed empirically.

2.4 Evaluation Metrics

2.4.1 Forecasting Accuracy: RMSE and MAE

Planned Content:

- Mathematical definition of root mean squared error highlighting its sensitivity to outlier predictions within point wise evaluation.
- Formulation of mean absolute error as a robust linear metric that provides direct clinical interpretation regarding average deviation magnitudes.

2.4.2 The lag_rmse Diagnostic Metric

Planned Content:

- Definition of the lag_rmse metric as the minimum possible residual error obtained after shifting the predicted sequence by an optimal constant time increment.
- Formulation of the shift penalty parameter as the difference between standard error and shifted error to quantitatively isolate temporal delay components.

2.4.3 Hypoglycemia Event Detection: F1, F2, and Tolerance τ

Planned Content:

- Definition of true hypoglycemia onset events determined by a distinct downward transition across the critical 70 mg/dL threshold boundary.

- Operational logic of time tolerance windows where a predicted event crossing is categorized as a true positive if it occurs within a set number of steps from a factual onset.
- Derivation of precision recall and harmonic F1 score metrics adapted to the evaluation of highly skewed event detection tasks.
- Motivation for the F2 score metric which applies a heavier weight to recall to prioritize clinical patient safety by penalizing missed conditions over false alarms.
- Description of Figure 2.4: Visual depiction of hypoglycemia detection showing true and predicted blood glucose levels where the tolerance parameter relaxes the specific crossing constraint.

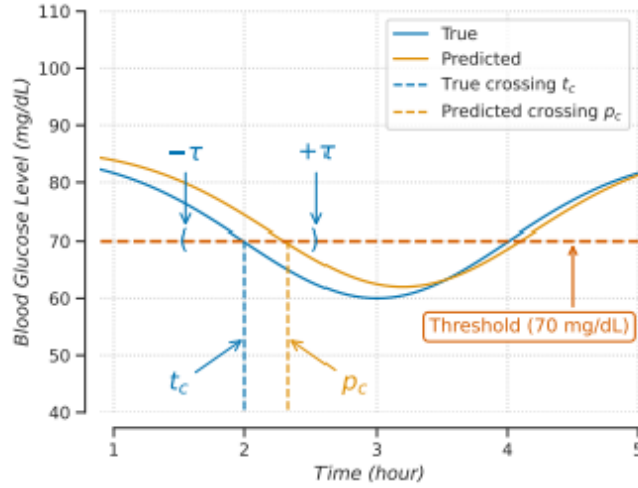


Figure 2.4: Tolerance-based matching of hypoglycemia events. The true crossing time t_c (blue dashed) and the predicted crossing time p_c (orange dashed) are counted as a True Positive match if $p_c \in [t_c - \tau, t_c + \tau]$. A larger tolerance τ relaxes the temporal matching constraint and is central to the τ -sweep evaluation introduced in Section 3.5.1. Adapted from [3].

2.5 Related Work

2.5.1 Glucose Forecasting with Deep Learning

Planned Content:

- Review of dominant recurrent and transformer-based neural structures applied within modern long-horizon metabolic time series prediction.
- Analysis of the trade-offs between univariate setups focused strictly on past tracking signals and multivariate configurations incorporating tracking logs for insulin and carbohydrates.
- Positioning of this evaluation framework against parallel deep learning architectures designed for continuous blood glucose prediction horizons.
- Description of Table 2.2: comparison of the most relevant prior approaches along the dimensions of architecture, input modality, dataset, prediction horizon, and reported error, with the present thesis included as the reference row.

Table 2.2: Comparison of representative deep-learning approaches to CGM-based glucose forecasting. The table is intended to position the present work within the broader landscape and to highlight that prior reports rarely separate shape error from temporal misalignment – a gap that the `lag_rmse` metric introduced in Section 2.4.2 explicitly addresses.

Work	Architecture	Input	Horizon	Reported metric	
Hüni [4]	LSTM / GAT	CGM only	15/30/60 min	F2 up to 0.50	
PRG proposal [3]	LSTM, PatchTST, NHITS, TimeLLM	CGM only	60 min	RMSE	36–41
TODO (e.g., Iacono et al.)	LSTM	multivariate	30 min	mg/dL	TODO
TODO (e.g., Feizlizardo et al.)	various	CGM + insulin + CHO	30/60 min	TODO	
This thesis	LSTM, PatchTST	CGM only	60 min	RMSE	+
				lag_rmse	+
				F1/F2	

2.5.2 Time-Shift Mitigation Strategies

Planned Content:

- Review of specialized alignment objective functions utilized across parallel speech recognition and sequence categorization research fields.

- Assessment of direct multi-step output prediction frameworks designed to preserve global phase alignment over expansive forecasting horizons.
- Examination of direct window-based binary classification pipelines configured as an alternative to thresholding continuous regression signals.

Chapter 3

Experimental Design and Training Objectives

This chapter describes the formal problem definitions, data partitioning strategies, hyperparameter optimization protocol, and the specific configurations of the three research objectives. Section 3.1 formalizes the prediction tasks across continuous regression and binary classification settings. Section 3.2 details data preparation protocols, model optimization pipelines, and the controlled ablation framework. Sections 3.3 through 3.5 define the mathematical criteria governing each independent objective.

3.1 Problem Formulation

Planned Content:

- Mathematical notation of the continuous glucose input sequence lookback window covering 24 sequential steps equivalent to 120 minutes of tracking.
- Formal definition of the single step forecasting target at a 60 minute horizon versus the complete multi step output trajectory vector.
- Formal presentation of the binary event classification label representing the presence of at least one hypoglycemia threshold breach within the forward window.
- Reiteration of the structural decision to optimize entirely decoupled per patient architectures without cross patient dataset mixing.

3.2 Experimental Setup

3.2.1 Dataset and Train/Val/Test Splits

Planned Content:

- Execution of the non-shuffled chronological partition into training (60 %), validation (20 %), and testing (20 %) fragments to preserve continuous temporal integrity.
- Justification for the chronological (rather than random) split: glucose values are strongly autocorrelated ($\rho_1 > 0.98$, see Section 2.1.1), so random shuffling would leak future information into the training set.
- Contrast between patient-wide standard normalization (used by the LSTM, computed on the training fragment only) and window-localized instance processing (RevIN, internal to PatchTST).
- Exclusion rule for individual patient profiles exhibiting zero positive hypoglycemia occurrences within the initial training fraction.
- Description of Figure 3.1: schematic illustration of the chronological partition per patient.

*Placeholder for `img/chronological_split.png`.
Schematic of the chronological 60/20/20 split per patient,
with time on the x-axis.*

Figure 3.1: Chronological 60/20/20 split applied independently to each patient’s CGM stream. The temporal ordering of the partitions is essential: because consecutive glucose readings are highly autocorrelated, any random shuffling would leak future information into the training set and inflate validation and test performance.

3.2.2 Model Architectures and Hyperparameter Optimization

Planned Content:

- Operational parameters of the Optuna automated search space framework executed exclusively on patient 85202 to lock final architectural settings.
- Configuration of the AdamW optimization engine combined with a rigorous early stopping patience threshold applied to validation loss.
- Design choice to freeze all architecture hyperparameters across every single objective to preserve strict ablation experimental control.

3.2.3 Ablation Design

Planned Content:

- Core isolation mechanism where a single network attribute transitions per objective block while maintaining completely symmetric baseline configurations.
- Structural baseline definition incorporating point wise single output estimation evaluated under conventional mean squared error criteria.
- Operational parameters for the modified loss function pipelines under Objective 1 alongside the multi step and binary classification adaptations.

3.3 Objective 1: Offset-Aware Loss Functions

3.3.1 Bounded-Lag Loss

Planned Content:

- Mathematical motivation for bounded lag formulations to alleviate point wise temporal penalty components during training optimization.

- Algorithmic step by step calculation of local minimum error evaluation over a specialized restricted step window configuration.
- Implications of setting the local neighborhood boundary parameter to exactly three steps representing 15 minutes of tracking tolerance.

3.3.2 Soft-DTW Loss

Planned Content:

- Algorithmic derivation of the differentiable soft dynamic time warping cost matrix leveraging recursive soft minimum paths.
- Analysis of regularized smoothing parameters set to unity to balance trajectory tracking stability with structural gradient optimization.
- Expected contrast regarding global elastic path distortions that accommodate severe macro delays at the cost of absolute level precision.

3.4 Objective 2: Multi-Step Forecasting

Planned Content:

- Implementation of the direct multi output framework extending the final projection dimension to encompass 12 simultaneous tracking horizons.
- Mathematical formula of the sequence averaged mean squared error criterion providing constant supervision across the entire predicted trajectory.
- Core hypothesis that continuous multi step training enforces strict phase coherence to structurally suppress the systematic time shift anomaly.

3.5 Objective 3: Event-Centric Evaluation

3.5.1 Tolerance-Sweep Evaluation

Planned Content:

- Evaluation design of the sequential tolerance sweep checking forecast threshold crossings across a continuous span of six distinct window settings.
- Analytical utility of mapping performance curves to determine the exact timestamp delay properties of independent regression models.

3.5.2 Direct Binary Event Classification

Planned Content:

- Reformulation of the operational objective moving from trajectory tracking to a direct forward window binary classification problem setting.
- Mathematical framework of the weighted binary cross entropy loss function incorporating patient specific inverse class frequencies.
- Specification of the fixed decision boundary set to 0.5 without subsequent operating point calibration to evaluate pure classification performance.

Chapter 4

Empirical Results of the Ablation Study

This chapter presents the empirical results of all experimental configurations evaluated across the cohort of 36 patients. Section 4.1 establishes the single-step baseline performance under mean squared error optimization. Section 4.2 reports the results of the offset-aware loss functions (Objective 1). Section 4.3 details the performance of the direct multi-step forecasting strategy (Objective 2). Section 4.4 transitions to an event-centric evaluation framework (Objective 3), incorporating the tolerance sweep, the direct binary classifier, and a lead-time analysis. Section 4.5 concludes with a comprehensive comparative summary across all methodologies.

4.1 Baseline: MSE-Trained Single-Step Forecasting

Planned Content:

- Direct descriptive reporting of the baseline results presented in Table 4.1, highlighting the lower point-wise RMSE of the recurrent architecture alongside the superior shape fidelity (lag_rmse) of the patch-based transformer.
- Description of the quantitative shift-penalty: the temporal misalignment component accounts for roughly 64% of total RMSE in PatchTST and 40% in the LSTM.

- Objective presentation of the event-centric detection metrics showing near-total failure for both baseline configurations at the standard 15-minute clinical tolerance level.
- Description of Figure 4.1: single-patient (85202) trajectory comparison illustrating that both predictions replicate the true shape but remain systematically delayed.
- Description of Figure 4.2: population-level histogram of per-patient best-lag Δ^* , demonstrating that the time-shift is systematic across the entire 36-patient cohort and not driven by individual outliers.

Table 4.1: Baseline single-step forecasting results, averaged over all 36 patients. RMSE and MAE are reported in mg/dL. The lag_rmse metric represents the point-wise error remaining after optimal constant time shifting; the Shift-Penalty equals $\text{RMSE} - \text{lag_rmse}$ and quantifies the share of total error caused purely by temporal misalignment. Hypo-F1 is computed at the clinical reference tolerance of 15 min ($\tau = 3$ steps). The mean best-lag is shown both in steps and in minutes for direct clinical interpretation.

Model	RMSE	MAE	lag_rmse	Shift-Penalty	Hypo-F1	Shift (steps)
LSTM	36.09	27.12	21.65	14.44	0.005	10.2 (51 min)
PatchTST	39.27	29.00	14.21	25.06	0.061	11.0 (55 min)

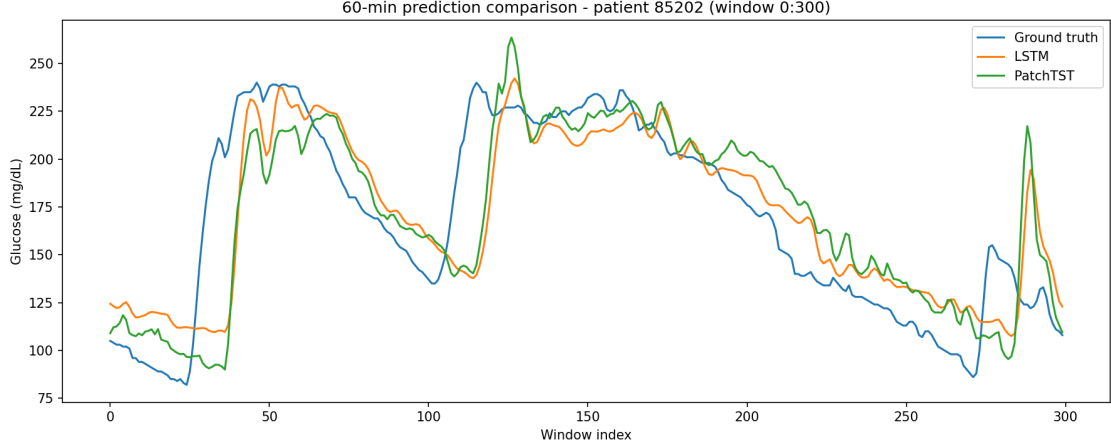


Figure 4.1: Baseline forecasting trajectories for the designated tuning patient 85202. The actual continuous glucose signal (blue) is contrasted against the LSTM (orange) and PatchTST (green) single-step predictions over a 60-minute prediction horizon. Both predictions reproduce the shape of the true trajectory but are visibly delayed – the visual expression of the systematic time-shift quantified by the Shift-Penalty column of Table 4.1.

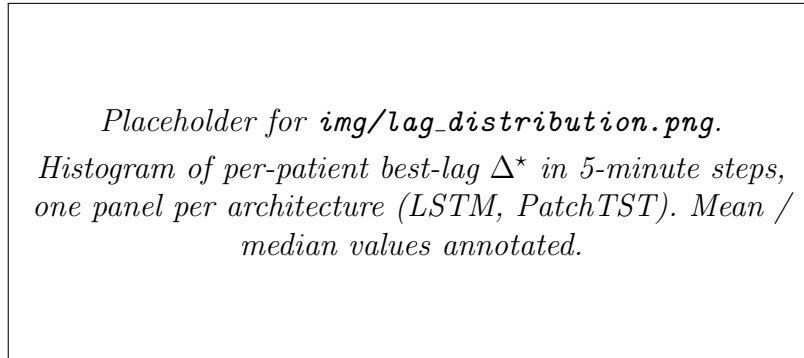


Figure 4.2: Distribution of the per-patient best-lag Δ^* obtained by maximising cross-correlation between predicted and true trajectories within a ± 12 -step window. Mass concentrated at large positive lags (around 10–11 steps, i.e. 50–55 min) demonstrates that the time-shift artefact is systematic across the cohort, not a property of individual patients. This figure is the population-level counterpart of the single-patient illustration in Figure 1.2.

4.2 Objective 1: Offset-Aware Loss Functions

4.2.1 Bounded-Lag Loss Results

Planned Content:

- Description of Table 4.2 regarding bounded lag optimizations noting that continuous trajectory errors remain nearly identical to the baseline while shape fidelity error metrics show a slight increase for both architectures.
- Statistical reporting of the marginal gains observed in event centric detection performance where both networks remain below a critical utility threshold.

4.2.2 Soft-DTW Loss Results

Planned Content:

- Description of the substantial point wise performance degradation shown in Table 4.2 under soft dynamic time warping optimization where error values increase sharply.
- Reporting of the expanded temporal alignment variance where elastic sequence matching during the training phase yields higher residual shape misalignment metrics.

4.2.3 Interpretation: Why Loss Relaxation Is Insufficient

Planned Content:

- Analytical interpretation of the overall objective results focusing on the core finding that relaxing temporal constraints does not supply the structural information required to anticipate future glycemic changes.
- Explanation of the fundamental tracking information deficit where univariate input structures lack access to the causal signals of exogenous insulin and carbohydrate intake.

Table 4.2: Evaluation of the offset-aware training criteria of Objective 1, averaged across the 36 patients of the T1DATA cohort. Baselines are repeated for ease of comparison. The Shift-Penalty column quantifies how much of the total RMSE remains attributable to temporal misalignment; values closer to zero indicate better phase alignment. Neither Bounded-Lag nor Soft-DTW reduces the Hypo-F1 gap to a clinically useful level, as discussed in Section 4.2.3.

Model	Loss Function	RMSE	lag_rmse	Shift-Penalty	Hypo-F1
LSTM	MSE (Baseline)	36.09	21.65	14.44	0.005
LSTM	Bounded-Lag ($D = 3$)	36.14	25.25	10.89	0.014
LSTM	Soft-DTW ($\gamma = 1$)	37.23	28.67	8.56	0.014
PatchTST	MSE (Baseline)	39.27	14.21	25.06	0.061
PatchTST	Bounded-Lag ($D = 3$)	39.30	16.70	22.60	0.067
PatchTST	Soft-DTW ($\gamma = 1$)	43.24	24.01	19.23	0.062

4.3 Objective 2: Multi-Step Forecasting

Planned Content:

- Description of the numeric values in Table 4.3 confirming that direct multi step sequence supervision is the solitary regression methodology that yields an absolute reduction in standard point wise error.
- Reporting of the architectural impact on the shape error metric where the recurrent model achieves its absolute lowest trajectory alignment error sequence.
- Interpretation of the multi step mechanism which demonstrates that simultaneous loss minimization across early steps acts as a temporal anchor to maintain phase coherence.

Table 4.3: Forecasting metrics for Objective 2 (multi-step direct forecasting), evaluated at the final 60-min horizon step ($h = 11$) so that the numbers are directly comparable to the single-step baselines. Multi-step training is the only regression configuration that simultaneously reduces RMSE *and* lag_rmse for both architectures, supporting the hypothesis that intermediate supervision acts as a temporal anchor.

Model	Output Strategy	RMSE	MAE	lag_rmse	Shift-Penalty	Hypo-F1
LSTM	Single-Step Baseline	36.09	27.12	21.65	14.44	0.005
LSTM	Multi-Step ($H = 12$)	35.59	26.80	20.74	14.85	0.010
PatchTST	Single-Step Baseline	39.27	29.00	14.21	25.06	0.061
PatchTST	Multi-Step ($H = 12$)	39.05	28.84	14.53	24.52	0.072

4.4 Objective 3: Event-Centric Evaluation

4.4.1 τ -Sweep: Event Detection Across Tolerance Levels

Planned Content:

- Description of Table 4.4 and Figure 4.3, tracking the F1 curves across tolerance ranges; note the flat response of recurrent configurations up to roughly the 20-minute boundary.
- Interpretation of the observed performance plateaus as direct evidence of the static ≈ 50 -minute prediction delay under standard single-step training.
- Brief pointer to the supplementary precision/recall breakdown in Figure B.1 (Appendix B), confirming that F1 gains at large τ are recall-driven.

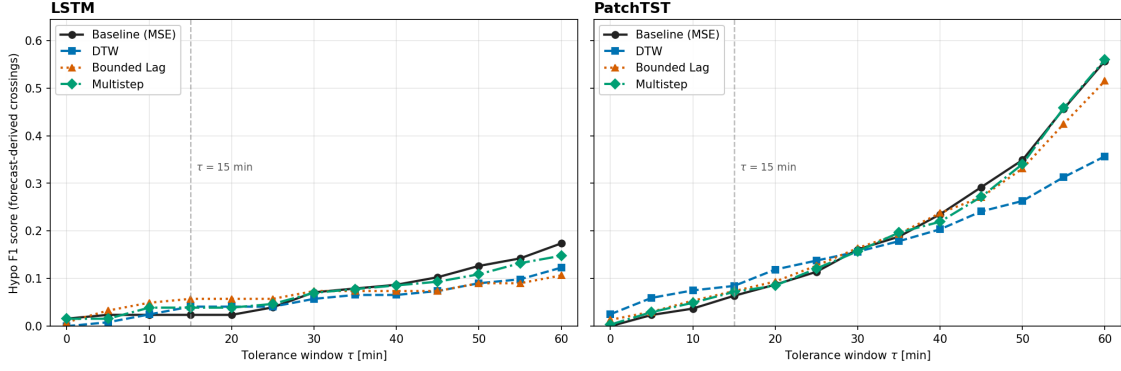


Figure 4.3: Hypo-F1 score of all forecast-derived event detectors as a function of the tolerance window τ , from 0 to 60 min, for LSTM (left) and PatchTST (right). The flat response of LSTM-based variants up to roughly $\tau = 20$ min is the empirical signature of a roughly 50-min static delay: only once τ grows large enough to absorb that delay do F1 scores begin to climb. The clinical reference $\tau = 15$ min is marked by the dashed vertical line.

Table 4.4: Hypo-F1 scores at six tolerance levels for all seven trained configurations. Column abbreviations: **L-** / **P-** refer to LSTM / PatchTST; **BL** = Bounded-Lag, **DTW** = Soft-DTW, **MS** = Multi-Step. The numbers underlie Figure 4.3 and the supplementary precision/recall breakdown in Figure B.1 (Appendix B); they provide direct comparison points between the four LSTM and three PatchTST training variants.

Tol. (τ)	LSTM	L-BL	L-DTW	L-MS	PTST	P-BL	P-MS
5 min	0.005	0.006	0.002	0.003	0.021	0.031	0.044
10 min	0.005	0.013	0.005	0.010	0.042	0.057	0.058
15 min	0.005	0.014	0.014	0.010	0.061	0.067	0.062
20 min	0.005	0.014	0.014	0.010	0.073	0.084	0.098
25 min	0.011	0.014	0.014	0.015	0.094	0.117	0.114
30 min	0.025	0.023	0.022	0.023	0.118	0.139	0.125

4.4.2 Direct Binary Event Classifier

Planned Content:

- Description of Table 4.5: direct classification performance, where the specialized LSTM classifier yields a 62-fold F1 improvement over its forecast-derived counterpart.
- Description of Figure 4.4 (qualitative output trace) and Figure 4.5 (precision-recall positioning); the aggregated TP/FP/FN block counts are provided in Figure B.2 (Appendix B).

- Analytical interpretation of the architecture performance inversion: the recurrent model retains absolute glucose level information, whereas the instance-normalized transformer forfeits the static threshold reference.
- Interpretation of Figure 4.5: precision, not recall, is the bottleneck for both direct classifiers.

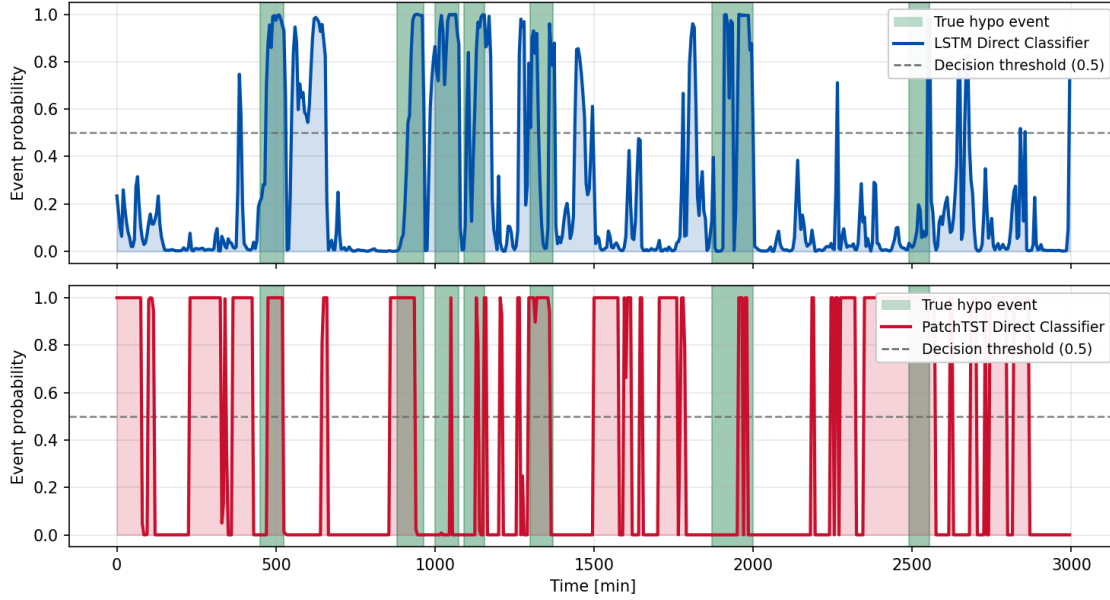


Figure 4.4: Direct event classifier output trace for a representative test patient. Green shaded blocks mark true hypoglycemia events; the decision threshold at 0.5 is shown by the dashed horizontal line. The LSTM classifier (top) produces concentrated peaks around true events, whereas the PatchTST classifier (bottom) produces wide, high-probability plateaus that frequently overlap non-event regions, a direct consequence of RevIN removing absolute glucose level information (cf. Section 2.2.2).

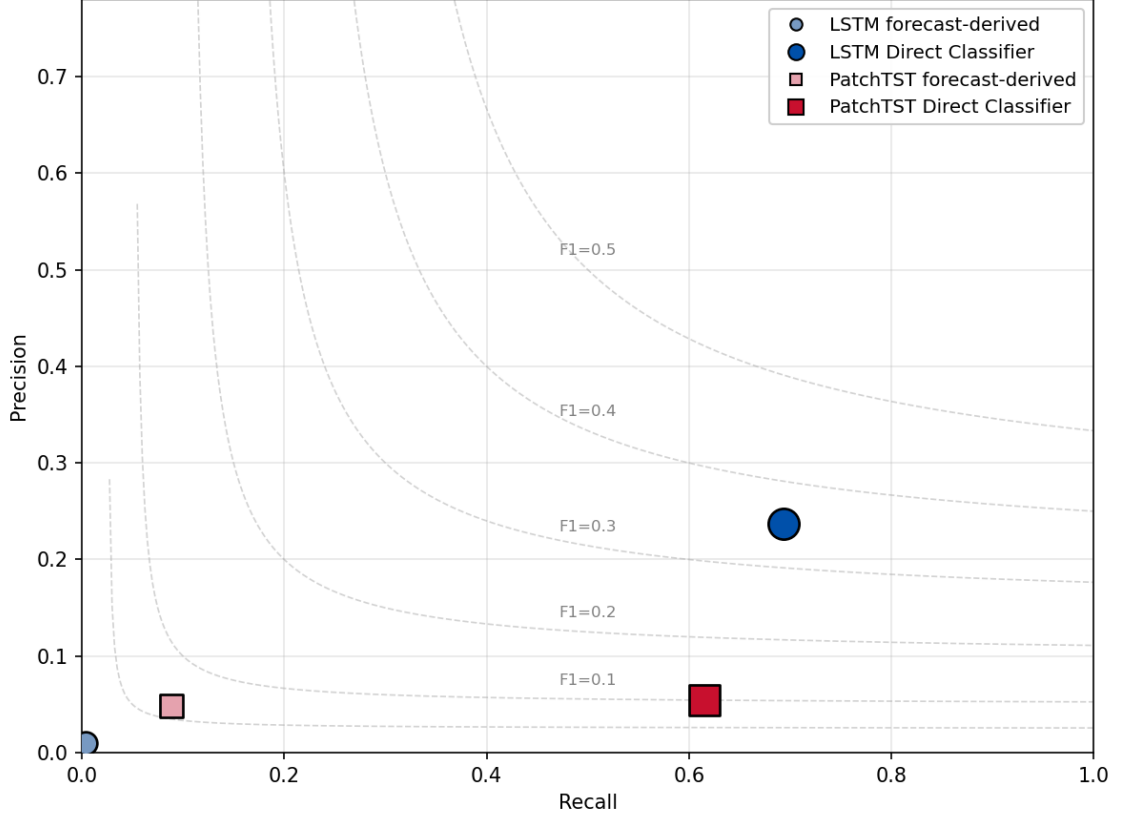


Figure 4.5: Precision-recall positions of all event detectors at the operational decision threshold of 0.5, with iso-F1 contours shown as dashed lines. Both direct classifiers achieve similar recall, but the LSTM classifier sits an entire iso-F1 contour higher in precision, explaining the factor-three difference in F1 despite comparable recall. The two forecast-derived points cluster near the origin and serve as a visual reference for the magnitude of the improvement.

Table 4.5: Performance comparison between the two direct binary classifiers and the best-performing forecast-derived detectors, all evaluated at the maximum 30-minute tolerance ($\tau = 6$ steps). The LSTM direct classifier achieves the highest F1 (0.334) of any configuration tested in this thesis – a 62-fold improvement over its forecast-derived counterpart ($F1 = 0.005$).

Model	Methodology	Precision	Recall	F1 Score	F2 Score
LSTM	Direct Classifier	0.237	0.693	0.334	0.463
PatchTST BL	Regression ($\tau = 30$)	0.227	0.139	0.176	0.177
PatchTST MS	Regression ($\tau = 30$)	0.220	0.138	0.163	0.178
PatchTST	Direct Classifier	0.054	0.615	0.095	0.147
LSTM Baseline	Regression ($\tau = 30$)	0.017	0.025	0.019	0.020

4.4.3 Lead-Time Analysis of the Direct Classifier

Planned Content:

- Description of Table 4.6: warning horizons of the two direct classifiers, noting the short median warning window achieved by the LSTM classifier despite its higher F1.
- Interpretation of the clinical utility limitations stemming from reactive tracking behaviors where alarms fire primarily during active threshold breaches rather than providing extensive advance notice.
- Brief pointer to the supplementary lead-time distribution in Figure B.3 (Appendix B), which shows the long-tailed shape behind the median values of Table 4.6.

Table 4.6: Lead-time statistics of the two direct binary event classifiers. The lead time is the duration between a true-positive alert and the actual threshold crossing. The LSTM classifier triggers earlier on average but also more often during the event itself (short median), whereas the PatchTST classifier produces a longer median warning window but at the cost of substantially lower precision (cf. Table 4.5).

Model Architecture	True Positives	Mean Lead	Median Lead
LSTM Event Classifier	2610	12.4 min	5.0 min
PatchTST Event Classifier	2243	18.3 min	15.0 min

4.5 Overall Comparison

Planned Content:

- Descriptive summary of the global metrics in Table 4.7, confirming that no individual architecture wins on both forecasting and event-detection tasks simultaneously.
- Final structural recommendation motivating a decoupled dual-model system that pairs multi-step trajectory regression with a specialized recurrent binary classifier for hypoglycemia early warning.

Table 4.7: Global evaluation matrix across all tested configurations of the ablation study. RMSE and lag_rmse are not defined for the direct classifiers, hence the dashes; Hypo-F1 and Hypo-F2 use the 30-minute tolerance ($\tau = 6$). The split between best forecasting (LSTM Multi-Step on lag_rmse) and best event detection (LSTM Direct Classifier on F1/F2) is the empirical basis for the dual-model recommendation made in Chapter 5.

Architecture	Optimization Methodology	RMSE	lag_rmse	Hypo-F1	Hypo-F2
LSTM	MSE Baseline	36.09	21.65	0.005	0.004
LSTM	Bounded-Lag ($D = 3$)	36.14	25.25	0.014	0.013
LSTM	Soft-DTW Loss	37.23	28.67	0.014	0.010
LSTM	Multi-Step Output	35.59	20.74	0.010	0.010
LSTM	Direct Binary Classifier	–	–	0.334	0.463
PatchTST	MSE Baseline	39.27	14.21	0.061	0.074
PatchTST	Bounded-Lag ($D = 3$)	39.30	16.70	0.067	0.080
PatchTST	Soft-DTW Loss	43.24	24.01	0.062	0.084
PatchTST	Multi-Step Output	39.05	14.53	0.072	0.085
PatchTST	Direct Binary Classifier	–	–	0.095	0.178

Chapter 5

Conclusions and Future Work

This chapter summarizes the central empirical findings of the ablation study discusses the underlying methodological limitations and outlines strategic directions for subsequent research. Section 5.1 aggregates the core insights from all three research objectives. Section 5.2 assesses the architectural and structural boundaries of the evaluated frameworks, and Section 5.3 establishes clear vectors for future investigative work.

5.1 Summary of Findings

Planned Content:

- Conclusion regarding the systematic time shift anomaly confirming that modifying the optimization criteria alone does not resolve the predictive delay.
- Identification of the fundamental tracking information deficit where univariate data structures lack access to exogenous causal factors.
- Summary of Objective 1 indicating that bounded lag and soft dynamic time warping criteria expand temporal variance without improving underlying shape alignment.
- Summary of Objective 2 establishing direct multi step sequence supervision as the superior methodology for continuous trajectory prediction.
- Summary of Objective 3 demonstrating the clear performance advantage of the specialized recurrent binary event classifier over standard regression thresholding.

5.2 Methodological Limitations

Planned Content:

- Acknowledgment of the strictly univariate constraint which mathematically bounds the predictive capacity during sudden metabolic fluctuations.
- Critical assessment of the short median advance warning window achieved by the primary classification architecture.
- Evaluation of the severe dataset class imbalance which restricts overall precision metrics and induces false alarm sequences.

5.3 Future Work

Planned Content:

- Proposal to integrate multimodal tracking sequences including logged insulin administration and carbohydrate consumption to eliminate the information deficit.
- Recommendation to evaluate expanded bounded lag optimization windows to fully encompass the observed systematic delay component.
- Suggestion to implement specialized focal loss functions to directly address minority class representation during neural network training.
- Perspective on transitioning to leave one out population configurations to measure cross patient generalizability.

Appendix A

Hyperparameter Details

Planned Content:

- Specification of the final locked architecture parameters derived from the designated tuning patient (Table A.1).
- Full description of the Optuna search spaces explored for both architectures (Table A.2).
- Detailed overview of the optimization configuration including the sampler (TPE), the pruning strategy (median pruner), and the total trial budget.
- Brief justification of the AdamW choice and early-stopping patience.

Table A.1: Final hyperparameters obtained after Optuna tuning on patient 85202 (50 trials, TPE sampler). These values remain constant across the baseline and all three objectives so that any observed difference is attributable to the change in training criterion, consistent with the controlled-ablation design.

Parameter	LSTM	PatchTST
hidden_size / d_model	256	64
num_layers	2	3
n_heads	–	4
ffn_dim	–	256
dropout	0.159	0.025
learning_rate	9.61×10^{-4}	1.21×10^{-4}
batch_size	128	128

Table A.2: Search space explored during Optuna hyperparameter optimization for both architectures. Search was performed on patient 85202 only; the resulting hyperparameters (Table A.1) were then applied unchanged to all 36 patients across all objectives. Categorical ranges are denoted $\{\cdot\}$ and continuous ranges $[\cdot, \cdot]$ on a log-uniform scale where appropriate.

Parameter	LSTM range	PatchTST range
hidden_size / d_model	$\{64, 128, 256\}$	$\{32, 64, 128\}$
num_layers	$\{1, 2, 3\}$	$\{2, 3, 4\}$
n_heads	–	$\{2, 4, 8\}$
ffn_dim	–	$\{128, 256, 512\}$
dropout	$[0.0, 0.3]$	$[0.0, 0.3]$
learning_rate	$[10^{-5}, 10^{-3}]$ (log)	$[10^{-5}, 10^{-3}]$ (log)
batch_size	$\{64, 128, 256\}$	$\{64, 128, 256\}$
sampler	TPE (Tree-structured Parzen Estimator)	
pruning strategy	Median pruner	
trial budget	50 trials per architecture	
objective metric	Validation MSE on patient 85202	

Appendix B

Supplementary Evaluation Figures

This appendix collects figures that support, but are not essential to, the main argument of Chapter 4. They are referenced from the relevant results sections and provide additional detail for the interested reader.

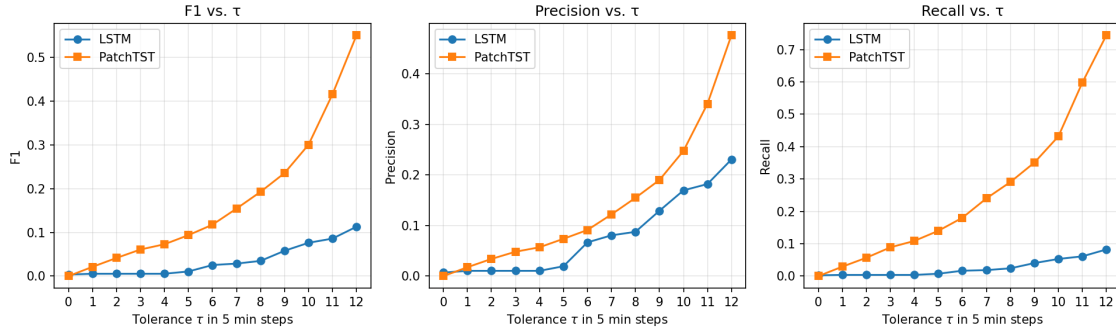


Figure B.1: Breakdown of the tolerance sweep into precision, recall, and F1 for the baseline LSTM and PatchTST configurations (cf. Section 4.4.1). Recall rises steeply with τ while precision lags, showing that the F1 gains at large τ are driven by relaxed timing constraints rather than by genuinely better predictions.

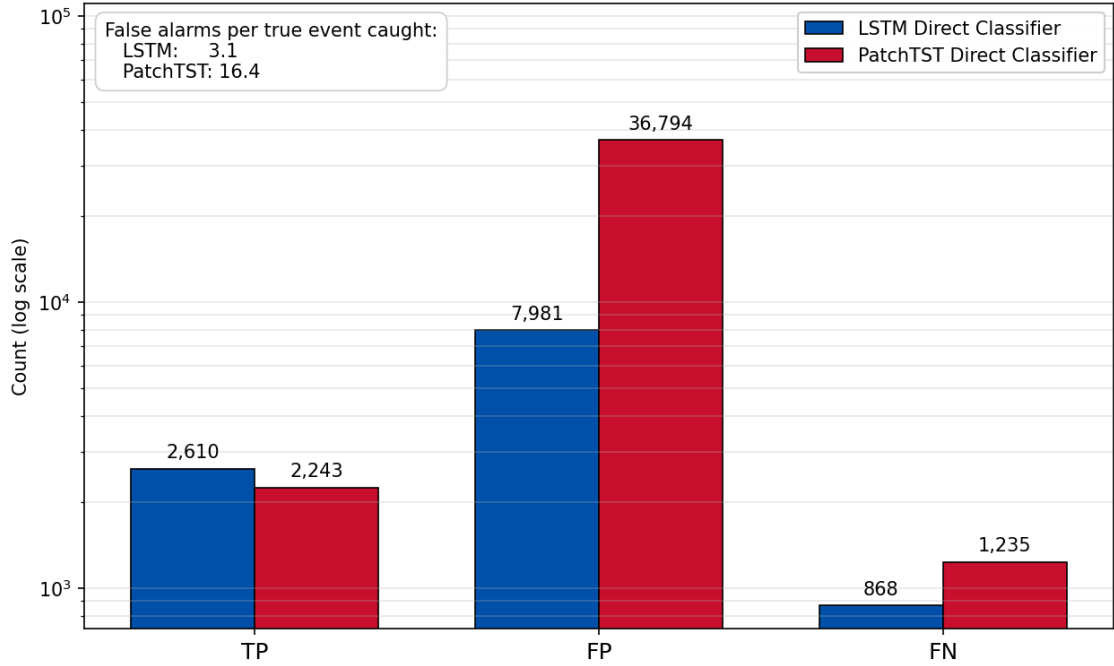


Figure B.2: Aggregated true positive, false positive, and false negative block counts over all 36 test patients on a logarithmic axis (cf. Section 4.4.2). The LSTM classifier produces 3.1 false alarms per true positive, whereas the PatchTST classifier produces 16.4 – the dominant source of the precision gap reported in Table 4.5.

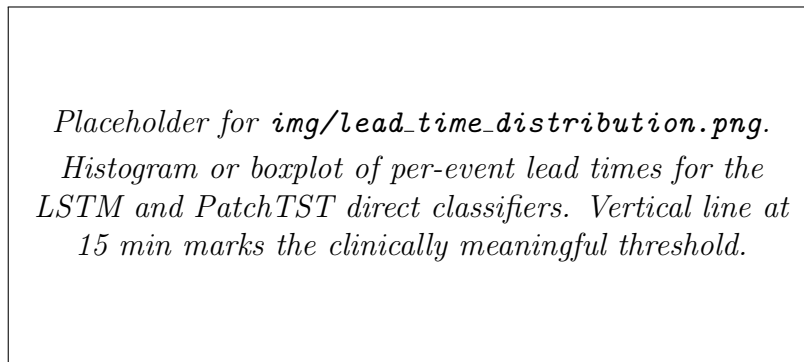


Figure B.3: Distribution of true-positive lead times for both direct classifiers (cf. Section 4.4.3). The median values of Table 4.6 hide a long-tailed distribution: a substantial fraction of LSTM alerts fires during the hypoglycemia event itself (lead time around 0 min), which is reactive rather than predictive.

Appendix C

Per-Patient Forecast Plots

Planned Content:

- Description of the supplementary forecast plot repository containing the comprehensive tracking evaluation profiles for the entire cohort.
- Presentation of representative tracking visualizations encompassing optimal alignment examples alongside edge cases exhibiting high variance.
- Note for the writing phase: do *not* embed all 36 plots individually. Show 2–4 representative examples (best/typical/worst alignment) here, and provide the full per-patient set as a compact multi-panel grid or as supplementary digital material, to keep the printed page count within budget.
- Optional: per-patient Δ^* table to complement the aggregated values reported in Section 4.1.

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